



Asset Pulse Application Help

Purpose

Asset Pulse is an oilfield forecasting and decision-support dashboard for evaluating well, pipeline, gathering-center and facility profitability under uncertainty. The application combines a React frontend with a Python FastAPI calculation backend and PostgreSQL/Neon storage.

Use it to:

- Build a monthly production and cash-flow case.
- Test CAPEX and OPEX overrun sensitivity.
- Estimate NPV, PV-10, payback, economic limit, EUR and netback.
- Compare exponential, hyperbolic and harmonic decline assumptions.
- Run tornado and Monte Carlo uncertainty analysis.
- Stack operational events such as downtime, price drops, OPEX escalation, restart cost and CAPEX overrun.
- Rank wells using a weighted shut-in/restart decision matrix.
- Export assumptions, cash flows and asset registers to CSV for Excel or database handoff.

Application Layout

Top Navigation

The top navigation switches between the main work areas:

View	Purpose
Scenario	Build and run one asset economic case. Includes the Upload data (CSV import) sub-panel so you can upload, approve, save and auto-run scenarios in one place.
Sensitivity	Run tornado sensitivity and Monte Carlo uncertainty. The header highlights which asset/scenario the tornado is currently running for.
Events	Apply event impacts to a base NPV case.
Decision Matrix	Rank assets for keep-online, choke-back, shut-in or restart decisions. Includes an in-app explainer covering each column and recommended bounds.
Assets	View seeded wells, pipelines, gathering systems and facilities. Includes a Refresh assets & cases button.
Scenario Compare	Compare every saved scenario by NPV, breakeven oil, payback and EUR, and download a PDF report of the comparison.



Note: the previous top-level *CSV Exchange* tab has been folded into the Scenario tab as the **Upload data** sub-panel — there is now a single place to upload, approve and save/run CSVs.

Left Command Rail

The command rail shows:

- Current loaded asset.
- Scenario horizon.
- CAPEX and OPEX multipliers.
- API status.
- Database backend currently used by the API.
- Active view.

Theme Toggle

Use [DARK] or [LIGHT] in the top-right corner to switch between light and dark mode. The theme uses React state only and does not store cookies or browser local storage.

Scenario View

The Scenario view is the main economic model. It also hosts the **Upload data** sub-panel for CSV import / approve / save & run, and its Load asset profile dropdown lists both base assets and saved scenarios/cases so you can hydrate inputs from any prior run.

Load Asset Profile dropdown

The dropdown groups options into:

- **Base assets** — wells, pads, pipelines, gathering systems and facilities

seeded in the database. Selecting one populates the form from the asset's cost profile (decline, OPEX, CAPEX rollup).

- **Saved scenarios / cases** — every scenario stored in the database, named

with the saved scenario name and its asset alias (e.g. Q1-A – asset1 [scenario #42]). Selecting one rehydrates the full input set from the saved JSON payload.

This means uploaded CSV cases (asset1, asset2, ...) appear in the dropdown the moment they are saved/run, so you can replay them from the Scenario tab without re-importing the file.

Running a Base Case

1. Open Scenario.
2. Select an asset profile from Load asset profile, or keep Custom well.
3. Review the key assumptions:
 - Horizon in months.
 - First-year decline.
 - Decline model.



- b-factor.
- Initial oil and gas rates.
- Oil price.
- Fixed OPEX.
- Development CAPEX.
- Discount rate.
- CAPEX and OPEX multipliers.

4. Keep `Apply economic limit truncation` enabled if you want the model to stop production once the well becomes uneconomic.
5. Click `Run scenario`.

Reset Button

The `Reset` button on the Scenario form restores every input — normal numeric fields, the fiscal regime, and all advanced/regime-specific fields — to the application defaults defined in `DEFAULT_INPUTS` (`frontend/src/components/ScenarioForm.tsx`). It also clears the `Load asset profile` selection so a previously loaded asset cannot silently re-hydrate the form. To work from an asset profile after `Reset`, pick the asset again from the dropdown.

Sample Scenario CSV

A starter template with five regime examples (US royalty/tax, NOC internal, PSC, TSC, concession/tax-royalty) is checked in at `examples/asset_pulse_scenario_input_template.csv`. Use it as the column reference when wiring up batch loads or external spreadsheets.

Key Inputs

Input	Meaning
Asset name	Label for the scenario report.
Horizon	Number of months to project.
First-year decline	Effective production decline during the first year.
Decline model	Exponential, hyperbolic or harmonic production decline.
b-factor	Arps hyperbolic b-factor; used only for hyperbolic/harmonic decline.
Initial oil	Starting oil production rate in BOPD.
Initial gas	Starting gas production rate in MCFD.
Oil price	Realized oil price in USD/bbl.
Fixed OPEX	Monthly fixed lease/facility cost.
Development CAPEX	Initial capital at month zero.
Discount rate	Annual discount rate for NPV.



Input	Meaning
CAPEX multiplier	Scenario factor for capital overrun or saving.
OPEX multiplier	Scenario factor for operating cost escalation or saving.

Advanced Inputs

Open Advanced `fiscal`, `fluid`, `OPEX` and `ARO` inputs to edit:

- NGL rate.
- Water cut start and end.
- Gas and NGL prices.
- Royalty.
- Production tax.
- Transport and processing cost.
- Variable oil/gas/water operating costs.
- Sustaining CAPEX.
- Abandonment/ARO cost.
- **Fiscal / Cost Regime** (see below).

Fiscal / Cost Regime

A single dropdown inside the advanced section controls how the base monthly cash flow is partitioned between contractor and host government. The default `us_royalty_tax` keeps existing behaviour (royalty and production tax already netted in revenue). Selecting any other regime swaps the contractor view into NPV, PV-10, payback, and the cash-flow chart. The full per-month breakdown is returned under `result.fiscal` and summarised in the analysis report.

Regime	What it does	Key inputs
<code>us_royalty_tax</code> (default)	No-op layer; reports the existing US royalty + production tax netback.	None.
<code>noc_internal</code>	Gross project economics with optional government transfer and corporate tax (default 0% / 0%) for NOC internal screening.	Gov transfer %, corp tax %.



Regime	What it does	Key inputs
psc_cost_recovery	PSC/EPSA: gross → royalty → available cost oil = (gross-royalty)×ceiling → actual cost oil = min(opex+capex×(1+uplift)+carry, available) → profit oil split (gov / contractor) → contractor tax. Unrecovered costs carry forward.	Royalty %, cost-oil ceiling %, contractor profit share %, contractor tax %, CAPEX uplift %.
technical_service_contract	TSC/RSC (Iraq-style): contractor reimbursed for eligible petroleum costs subject to a periodic cap (share of gross revenue) plus a flat remuneration \$/BOE. Carry-forward for unpaid eligible costs. No hydrocarbon ownership.	Payment cap % of revenue, remuneration \$/BOE, contractor tax %.
concession_tax_royalty	Middle East concession with royalty (flat or progressive) plus upstream income tax. Optional Saudi-style progressive royalty: 15% on the first \$70/bbl, 45% from \$70-\$100, 80% above \$100; income tax 50% (AGSI).	Flat royalty % or progressive toggle, income tax %.

References: Bindemann/Oxford Energy WPM 25 (1999); Daily Jus on Omani EPSA disputes (2024); PMC fiscal-regimes review (PMC7798991); AGSI Aramco analysis.

> **Need help populating the advanced CAPEX/OPEX/fiscal fields?** See the > step-by-step calculation guide at > [docs/help-me-calculation-guide.pplx.md](help-me-calculation-guide.pplx.md) > — it explains every field (fixed_opex_usd_month, variable_opex_usd_boe, > water_handling_usd_bbl, water_cut_pct, capex_multiplier, > opex_multiplier, downtime_months, downtime_start_month, > psc_royalty_rate_pct, psc_cost_recovery_ceiling_pct, > psc_contractor_profit_oil_share_pct, contractor_tax_rate_pct, > capex_uplift_pct, tsc_fee_usd_boe, tsc_cost_recovery_cap_usd_month, > concession_royalty_rate_pct, concession_income_tax_rate_pct) with > formulas, QA checks and worked NOC / IOC / US Independent examples.

Outputs

The KPI strip and analysis report show:

- NPV.
- PV-10.
- Payback.
- Netback/BOE.
- EUR in BOE.
- Economic limit.



- Truncation status.
- Discount rate.
- Decline model.
- First 12 months of projected cash flow.
- Profitability commentary.

The cash-flow projection panel displays monthly free cash flow and cumulative free cash flow.

Sensitivity View

The Sensitivity view contains two uncertainty tools. The header on each panel **explicitly highlights which asset/scenario the analysis is currently running for** — e.g. Tornado sensitivity – Eagle Ford Pad-3 well A plus a coloured caption that reads Tornado sensitivity for: Eagle Ford Pad-3 Well A . scenario #42 when a saved scenario is loaded.

Tornado Sensitivity

Click Run tornado sensitivity to test one-variable-at-a-time NPV swings.

Default drivers include:

- Oil price.
- Initial oil rate.
- Annual decline.
- CAPEX multiplier.
- OPEX multiplier.
- Discount rate.
- Final water cut.

Why run a tornado sensitivity? It varies one assumption at a time by a fixed $\pm\%$ while holding everything else constant, then ranks each input by the resulting NPV swing. The longest bar is the dominant driver. This achieves:

- Identification of the **key economic levers** so capital and OPEX attention

go to inputs that actually move NPV.

- Surfacing of **risk controls** — large downside swings flag candidates for

hedging, contracting, or tighter operational monitoring.

- A quick **sanity check** before running Monte Carlo: drivers that don't move

NPV here rarely justify modelling their full distribution.

Use this view to identify the biggest value driver before running a broader Monte Carlo simulation.

Monte Carlo

Click Run Monte Carlo to simulate NPV uncertainty.



Default uncertainty distributions:

- Oil price triangular distribution.
- CAPEX multiplier triangular distribution.
- OPEX multiplier triangular distribution.

Outputs:

- P10 NPV.
- P50 NPV.
- P90 NPV.
- Mean NPV.
- Standard deviation.
- Minimum and maximum simulated NPV.
- Downside risk commentary.

Use P10 as the downside case, P50 as the median case and P90 as the upside case.

Events View

The Events view stacks operational or commercial shocks on top of the current scenario result.

Typical event types:

- CAPEX overrun.
- Downtime.
- Price drop.
- OPEX escalation.
- Restart cost.

Workflow:

1. Run a scenario first.
2. Open Events.
3. Add one or more events.
4. Review each event's NPV impact.
5. Review the final adjusted NPV.

Use this view for “what happens if” analysis after the base case is already understood.

Decision Matrix View

The Decision Matrix ranks wells or assets for operational decisions. The view opens with an **in-app explainer** describing the purpose of the weighted matrix, the meaning of every column, and the recommended bounds for each.



What it does

A weighted decision matrix ranks wells/assets across operational and economic criteria to support **keep-online · choke-back · shut-in · restart** calls. Each criterion is normalised across the asset population (0–1), oriented so higher = better, then multiplied by its weight. The weighted score is the sum; a higher score favours keeping the asset online, while a high *shut-in pressure* (driven by avoidable OPEX and water burden) flags candidates to curtail.

Columns and recommended bounds

Column	Meaning	Direction	Recommended bounds
Asset	Well/pad name.	—	free text
Monthly margin (USD)	Net monthly cash margin at current prices.	higher better	typical –50,000 to +250,000 USD/mo
NPV keep online (USD)	Discounted NPV if the asset keeps producing through its economic life.	higher better	population-relative (no hard cap)
Avoidable OPEX (USD/mo)	Fixed OPEX that disappears when shut in.	raises shut-in pressure	0–50,000 USD/mo typical
Restart payback (mo)	Months to recover restart cost.	lower better	1–60 months
Restart risk (0–1)	Probability the well does not restart cleanly.	lower better	0.0–1.0
HBP / lease / midstream risk (0–1)	Risk of losing HBP, lease commitments or pipeline nominations.	lower better	0.0–1.0
Water burden (0–1)	Relative water-handling/disposal load.	raises shut-in pressure	0.0–1.0
Strategic value (0–1)	Non-economic value (data, partner, regulator, reservoir).	higher better	0.0–1.0

Bounds to respect

- 0–1 fields must stay between **0.0** and **1.0**.
- Default weights sum to **1.00** (20/15/10/10/15/10/10/10) — keep custom

weights on a 0–1 scale and summing to ~1.

- Currency fields have no enforced cap; the matrix re-normalises them across

the asset population on every score, so a single outlier compresses the others.

Click [Score matrix](#) to recompute recommendations after editing assumptions.

Use this view when production curtailment or restart choices require more than a simple monthly margin test.



Assets View

The Assets view lists seeded sample assets:

- Wells.
- Pipelines.
- Facilities.
- Gathering-center style assets.

Click **load** to push an asset profile into the Scenario form.

Refresh assets & cases

The view exposes a **Refresh assets & cases** button. The app automatically re-fetches `GET /api/assets` and `GET /api/scenarios?limit=100` whenever:

- A scenario is run (Scenario → Run scenario).
- One or more rows are saved/run via the Scenario tab's Upload data sub-panel.
- A scenario is deleted from Scenario Compare.

The same refresh feeds the **Load asset profile** dropdown in the Scenario tab, so newly imported cases (e.g. asset1, asset2) become loadable without a page reload. **Refresh assets & cases** is the manual fallback if an auto-refresh missed a change because of a network blip.

In production, replace the seeded sample data with your actual AFE, LOE, production, water, facility and midstream data.

Scenario Compare View

The Scenario Compare view is a portfolio-level table and chart of every saved scenario in the database. It loads from `GET /api/scenarios?limit=100` and shows:

- Scenario name and asset alias (e.g. asset1, asset2).
- Fiscal regime and source (api, csv_import, manual).
- NPV (positive in green, negative in red), PV-10 and payback.
- **Breakeven oil price** — the bisection-derived USD/bbl that drives NPV to ~zero, holding all other inputs constant. The KPI is computed once when the scenario is saved and cached on `scenario_results.breakeven_oil_price`.
- Netback per BOE and EUR (BOE).

Below the table, two simple SVG-style bars rank the cases by:

1. NPV (signed, with a midline).
2. Breakeven oil price (lower bar = more resilient case).

A **Refresh** button reloads the saved list. A free-text filter narrows the view by name, asset, regime or source.

Sort orders by newest, NPV, breakeven or payback. The **delete** action on each row removes a saved scenario from the database.

Scenarios appear here automatically when:



- You click **Run scenario** on the Scenario tab — runs persist by default.
- You click **Save approved** or **Save & Run** in the Upload data sub-panel

on the Scenario tab.

Generate PDF report

A **Generate PDF report** button at the top of the Scenario Compare toolbar calls POST `/api/scenarios/report.pdf` with the IDs currently shown after filtering and sorting. The backend renders a multi-page PDF using ReportLab that contains:

- Title Asset Pulse Scenario Comparison Report and a UTC generation

timestamp.

- Short interpretation bullets (best/worst NPV, most resilient breakeven,

fastest payback).

- A comparison table with NPV, PV-10, payback, breakeven oil and the fiscal

regime for every scenario.

- Mono-spaced bar charts of NPV (signed), breakeven oil price and payback

months — no external image dependencies, so the PDF renders identically on every Docker host.

- A per-scenario detail page with KPIs and the key inputs (horizon, oil rate,

decline, oil price, dev CAPEX, discount rate).

The endpoint streams application/pdf with a Content-Disposition: attachment header. The frontend names the download `asset-pulse-scenario-report-<timestamp>.pdf`. The report is generated from your saved scenarios only — it does not include external citations.

Upload data — CSV import sub-panel (inside Scenario tab)

The CSV import workflow lives directly under the Scenario inputs / report on the Scenario tab. The flow is:

1. Click **Import scenario CSV** and pick a one-row export or the multi-row

template at `examples/asset_pulse_scenario_input_template.csv`.

2. Approve the rows by ticking checkboxes (defaults to all).

3. Click **Save & Run** — the backend persists each row and runs the

economics, returning NPV and breakeven oil per row.

4. As soon as the request completes, the Asset registry, the Scenario Compare

tab, and the **Load asset profile** dropdown are refreshed automatically. The **first** saved row is also rerun to fetch its full monthly cash flow, which is then projected into the visible Scenario report panel and chart so you immediately see economics for what you imported. The form is bound to that scenario row, so a follow-up **Run scenario** updates it in place.

5. The CSV file picker shows Importing... and is disabled while parsing,



then re-enables as soon as rows are listed. Per-row parse errors do not freeze the screen — they appear inline as validation notes on the row.

6. Upload data tab is now part of the Scenario section, exactly as the

workflow expects: upload → approve → save → auto-run → loadable from the dropdown.

The legacy top-level *CSV Exchange* tab has been removed; everything below is how this sub-panel behaves.

Export Scenario Inputs

Click `Export scenario inputs` to download a one-row CSV containing current scenario assumptions.

Use this when:

- You want to review assumptions in Excel.
- You want to create scenario templates.
- You want to send assumptions to another engineer for review.

Import Scenario CSV

Use `Import scenario CSV` to upload either:

- a one-row export from `Export scenario inputs`, or
- the multi-row template at `examples/asset_pulse_scenario_input_template.csv`.

After import, every row in the file is shown in an **Imported scenarios** table with scenario name, asset, regime and key economics. Click `Load to Scenario` on the row you want to evaluate. The app:

1. Pushes the parsed inputs into the Scenario form (overwriting current values).
2. Clears any stale result so the next `Run scenario` reflects the new case.
3. Switches to the **Scenario** tab automatically so you can review and run.

Approve & Save Imported Rows

Each parsed row also gets a checkbox. Tick the rows you want to keep, then:

- **Save approved** — persists the inputs to the database without running

economics. Useful for staging future scenarios.

- **Save & Run** — persists each row, runs `project_scenario` server-side,

and stores the resulting NPV, payback, breakeven oil price and monthly summary on `scenario_results`. The saved IDs are echoed back in the feedback line; the rows then appear in the **Scenario Compare** tab and in the Assets view (after a refresh).

Behind the scenes the UI POSTs to `/api/scenarios/import` with one payload containing all selected rows. Each row stores `asset_id_or_name` as `scenarios.asset_alias`. If that alias matches an existing asset by id or case-insensitive name, the foreign key `scenarios.asset_id` is also set — otherwise only the alias is kept and the existing asset table is left untouched, so naming new cases like `asset1`, `asset2` etc. is safe.

Rules:

- Column names are case- and underscore-insensitive. Both raw API field



names (e.g. `oil_price`) and template column names (e.g. `oil_price_usd_bbl`) are recognised.

- Percentage columns in the template (e.g. `decline_rate_annual_pct = 45`,

`royalty_rate_pct = 18.75`) are converted to the fractions the API expects (`0.45`, `0.1875`).

- `scenario_name`, `asset_id_or_name` and notes are kept as display

metadata; the form's **Asset name** field falls back to `asset_id_or_name` when no `asset_name` column is present.

- `apply_economic_limit` and `concession_royalty_progressive` accept

`true/false/yes/no/1/0`.

- Blank fiscal/regime cells are ignored — the regime defaults stay in place.
- Invalid numeric or boolean cells are listed under **Validation notes**

beneath the imported table; the rest of the row still loads.

- Columns the app does not recognise (e.g. `working_interest_pct`,

`downtime_months`) are ignored without error.

Export Cash Flow

After running a scenario, click `Export cash flow` to download monthly results.

The export includes:

- Month.
- Net revenue.
- OPEX.
- Sustaining CAPEX.
- Development CAPEX.
- Abandonment.
- Free cash flow.
- Oil, gas and water volumes.

Export Asset Register

Click `Export asset register` to export all currently loaded assets and compact JSON assumptions.

Recommended Workflow

1. Load an asset or build a custom well case.
2. Run Scenario and inspect KPIs. The result is automatically saved.
3. Switch the decline model and compare exponential vs hyperbolic economics.
4. Run Tornado to identify dominant drivers.
5. Run Monte Carlo to quantify distribution risk.
6. Add Events to test operational shocks.



7. Use Decision Matrix for keep-online, choke-back, shut-in or restart calls.
8. Export scenario and cash flow CSVs for documentation.

Portfolio Workflow (CSV Import → Compare)

1. Build a multi-row CSV from `examples/asset_pulse_scenario_input_template.csv`.

Use distinct asset aliases such as `asset1`, `asset2`, ... in the `asset_id_or_name` column.

2. Open **CSV Exchange**, click `Import scenario CSV`.
3. Review the parsed rows, untick anything you do not want stored, then click

Save & Run.

4. Open **Scenario Compare** to inspect NPV, breakeven oil price and payback

side-by-side; sort or filter to spot the best case.

5. Open **Assets** and click **Refresh cases** if the new scenarios are not already visible.

9. Persist or seed production data in Neon once validated.

Interpretation Notes

- NPV is sensitive to price, production decline, water handling cost and fixed OPEX.
- Economic limit truncation is useful for marginal wells, but lease obligations, HBP status, restart risk and strategic value must still be considered.
- Monte Carlo ranges are only as good as the distribution assumptions.
- Hyperbolic decline preserves more tail production than exponential decline for the same first-year decline assumption.
- The tool is a decision-support model, not a substitute for final reserves, accounting, tax or regulatory review.

Troubleshooting

Issue	Likely Cause	Fix
API unreachable	FastAPI backend is not running or CORS is wrong.	Check backend host, <code>/api/health</code> , and <code>CORS_ORIGINS</code> .
Database shows SQLite fallback	<code>DATABASE_URL</code> is missing on backend host.	Set Neon pooled <code>DATABASE_URL</code> and restart backend.
Netlify page loads but calculations fail	Frontend cannot reach FastAPI.	Set <code>VITE_API_BASE_URL</code> or configure Netlify proxy redirect.
Scenario import shows no rows	File has only a header line, or commas inside cells are unquoted.	Verify at least one data row and quote any cell containing a comma.



Issue	Likely Cause	Fix
Loaded row values look wrong by 100×	Columns named *_pct were already entered as fractions instead of percentages.	The template treats *_pct columns as percentages — enter 45 for 45%, not 0.45.
Imported numbers ignored	Cell contains text or units the parser can't strip (e.g. 45 percent).	Use plain numbers; \$, , , _ , spaces, and a trailing % are stripped automatically.
Monte Carlo is slow	Iterations are high or backend host is cold-starting.	Start with 500–1000 iterations; increase after deployment is stable.
Results look too optimistic	Decline, water cut, OPEX or abandonment assumptions may be too low.	Test downside cases in Tornado and Monte Carlo.

Production Readiness Checklist

- Replace seeded sample data with operator-specific data.
- Connect backend to Neon using pooled DATABASE_URL.
- Set Netlify VITE_API_BASE_URL to the deployed FastAPI URL.
- Set backend CORS_ORIGINS to the Netlify domain.
- Add authentication before exposing internal economics publicly.
- Add role-based data access for multi-user production use.
- Validate fiscal terms and tax logic for the operating region.
- Back up Neon data before bulk imports.